David Saykin

Curriculum Vitae

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Education

2013–2017 **Bachelor of Science**, *Moscow Institute of Physics and Technology* (MIPT), Department of General and Applied Physics, **GPA** — 8.97 out of 10.
Specialization: Applied Mathematics and Physics
Chair: Problems in Theoretical Physics @ *Landau Institute for Theoretical Physics*

Bachelor thesis

Title **Magnetoconductance of p–n junction in Weyl semimetal**.

Supervisor Konstantin S. Tikhonov, Ph.D.

Description We theoretically study the magnetocondactance $G(B)$ of the p–n juction realized in ballistic Weyl semimetal in two–cone model. We show that unlike in the case of one–cone model where Klein tunneling manifests itself in positive differential magnetoconductance $dG(B)/dB$, the physically realistic two–cone model leads to the sign change in the magnetoconductance in magnetic fields of order $B_c \approx \Phi_0 k_0^2$, where k_0 is a cone splitting in momentum space and Φ_0 is a flux quantum.

Experience

Vocational

Autumn 2013 – Spring 2014 **Lab Assistant**, *Laboratory of Applied Computational Geophysics*, MIPT.
Half–year research on the inverse problem of wave diffusion in solids.
Supervisor: Vasily I. Golubev

Miscellaneous

Summer 2013 **Physics Teacher**, *Summer School «Quant»*, Kazan.
One mounth intesive teaching of Olympiad Physics to talanted pupils at summer school «Quant» organized by Kazan Federal (State) University.

Languages

Russian	Native	*Mother Tounge*
English	Advanced	***TOEFL**: 102*
German	Intermediate	***B2**–level 2–year course @MIPT*

Honors & Awards

2010, 2011 **Twofold Awardee**, *Physics Olympiad of Russia*, Republic Stage.

Computer skills

Operation systems	GNU/Linux (Ubuntu, Debain), Windows	Programming Languages	C, Assembler, Python, Java
Numerical Analysis Software	Wolfram Mathematica, GNU-plot, MATLAB, NumPy	Version Control Systems	Git (GitHub), SVN
Office	LibreOffice, Microsoft Office, Gimp, Inkscape, GeoGebra	Edition Editors	LaTeX 2_ε, XeTeX, TexStudio, Lyx, Vim

Interests & Hobbys

Sports	Martial Arts (Aikido, Jiu Jitsu), Skiing, Track and Field, Bicycling, Hiking, Camping (Head scout in «Space Scouts Organization»)
Music	Guitar, Ukulele, Vocal, Flute

References

Advisor/Teacher

Konstantin S. Tikhonov, Ph.D.
Researcher @ LITP, Russia
Email: tikhonov@itp.ac.ru

Teacher/Head of Chair

Mikhail V. Feigel'man, Sc.D.
Deputy Director @ LITP, Russia
Email: feigel@landau.ac.ru